$$\begin{aligned}
2\cos\frac{a+b}{2}\cos\frac{a-b}{2} &= 2\left(\cos\frac{a}{2}\cos\frac{b}{2}-\sin\frac{a}{2}\sin\frac{b}{2}\right)\left(\cos\frac{a}{2}\cos\frac{b}{2}+\sin\frac{a}{2}\sin\frac{b}{2}\right)\\
&= 2\left(\cos^2\frac{a}{2}\cos^2\frac{b}{2}-\sin^2\frac{a}{2}\sin^2\frac{b}{2}\right)\\
&= 2\left(\frac{1+\cos a}{2}\cdot\frac{1+\cos b}{2}-\frac{1-\cos a}{2}\cdot\frac{1-\cos b}{2}\right)\\
&= \cos a+\cos b
\end{aligned}$$

$$\therefore \cos(4a)+\cos(2a) = 2\cos 3a\cos a$$

Similarly,

$$2\sin\frac{a+b}{2}\cos\frac{a-b}{2} = \sin a+\sin b$$

$$\therefore \sin(4a)-\sin(2a) = 2\sin a\cos 3a$$

(i)

$$\begin{aligned}
4\cos x\cos 2x\cos 3x &= 2\cos 2x(\cos 4x+\cos 2x)\\
&= 2\cos 2x\cos 4x+2\cos^2 2x\\
&= \cos 6x+\cos 2x+1+\cos 4x
\end{aligned}$$

$$\begin{aligned}
\cos 6x+\cos 4x+\cos 2x &= 0\\
2\cos 4x\cos 2x+\cos 4x &= 0\\
\cos 4x(2\cos 2x+1) &= 0
\end{aligned}$$

$$\cos 4x = 0 \quad \text{or} \quad \cos 2x = -\frac{1}{2}$$

$$x = \frac{\pi}{8},\frac{3\pi}{8},\frac{5\pi}{8},\frac{7\pi}{8} \quad \text{or} \quad x = \frac{\pi}{3},\frac{2\pi}{3}$$

(ii)

$$\begin{aligned}
\tan x &= \tan 2x \tan 3x \tan 4x \\
\sin x \cos 2x \cos 3x \cos 4x &= \cos x \sin 2x \sin 3x \sin 4x \\
\frac{1}{2}(\sin 4x - \sin 2x) \cos 2x \cos 4x &= \frac{1}{2}(\sin 4x + \sin 2x) \sin 2x \sin 4x \\
\sin 4x(\cos 2x \cos 4x - \sin 2x \sin 4x) &= \sin 2x(\cos 2x \cos 4x + \sin 2x \sin 4x) \\
\sin 4x \cos 6x &= \sin 2x \cos 2x \\
\sin 4x \left(\cos 6x - \frac{1}{2}\right) &= 0 \\
\cos 6x = \frac{1}{2} \quad \text{or} \quad \sin 4x &= 0 \\
x = \frac{\pi}{18}, \frac{5\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18} \text{ or } x &= 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi
\end{aligned}$$

$$x = \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4} \text{ is not defined for } \tan x$$

$$\therefore x = 0, \frac{\pi}{18}, \frac{5\pi}{18}, \frac{7\pi}{18}, \frac{11\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18}, \pi$$

This solution was submitted by **Jon**

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